HBase Assignment

Beginner & Intermediate

Individual Practice Task

Overview

In this assignment you will build and query a product inventory system using HBase. You will work with a table called store that holds information about products across two column families.

Table Name	Column Families	Total Records
store	details, inventory	5 products

Dataset — store Table

Column Family	Columns
details	name, brand, category
inventory	price, quantity, warehouse

Row Key	Name	Brand	Category	Price	Quantity	Warehouse
1	laptop	dell	electronics	1200	50	A
2	phone	samsung	electronics	800	120	A
3	desk	ikea	furniture	300	30	B
4	chair	ikea	furniture	150	60	B
5	tablet	apple	electronics	600	80	A

Instructions

1. Read each question carefully before writing your command.
2. Write your HBase shell command(s) in the answer box below each question.
3. Test every command in your HBase shell and verify the output before submitting.
4. The dataset above is your only reference — do not use the emp or student tables.

Answers are provided at the end of this document. Do not look ahead until you have attempted all questions.

PART A — BEGINNER

Questions 1–6 cover table creation, inserting data, scanning, getting, and filtering.

Q1 Create the store table with two column families: details and inventory.

Your Answer:

```
hbase(main):014:0> create 'store', 'details', 'inventory'
Created table store
Took 0.7061 seconds
=> Hbase::Table - store
```

Q2 Insert all 5 rows from the dataset above into the store table. Insert all columns for both column families.

```
hbase(main):015:0> put 'store','1','details:name','laptop'
Took 0.0393 seconds
hbase(main):016:0> put 'store','1','details:brand','dell'
Took 0.0059 seconds
hbase(main):017:0> put 'store','1','details:category','electronics'
Took 0.0041 seconds
hbase(main):018:0> put 'store','1','inventory:price','1200'
Took 0.0065 seconds
hbase(main):019:0> put 'store','1','inventory:quantity','50'
Took 0.0043 seconds
hbase(main):020:0> put 'store','1','inventory:warehouse','A'
Took 0.0084 seconds
hbase(main):021:0>
hbase(main):022:0* put 'store','2','details:name','phone'
Took 0.0040 seconds
hbase(main):023:0> put 'store','2','details:brand','samsung'
Took 0.0065 seconds
hbase(main):024:0> put 'store','2','details:category','electronics'
Took 0.0063 seconds
hbase(main):025:0> put 'store','2','inventory:price','800'
Took 0.0050 seconds
hbase(main):026:0> put 'store','2','inventory:quantity','120'
Took 0.0067 seconds
hbase(main):027:0> put 'store','2','inventory:warehouse','A'
```

```
bigdata@localhost:~
File Edit View Search Terminal Help
hbase(main):036:0* put 'store','4','details:name','chair'
Took 0.0049 seconds
hbase(main):037:0> put 'store','4','details:brand','ikea'
Took 0.0043 seconds
hbase(main):038:0> put 'store','4','details:category','furniture'
Took 0.0044 seconds
hbase(main):039:0> put 'store','4','inventory:price','150'
Took 0.0065 seconds
hbase(main):040:0> put 'store','4','inventory:quantity','60'
Took 0.0096 seconds
hbase(main):041:0> put 'store','4','inventory:warehouse','B'
Took 0.0079 seconds
hbase(main):042:0>
hbase(main):043:0* put 'store','5','details:name','tablet'
Took 0.0074 seconds
hbase(main):044:0> put 'store','5','details:brand','apple'
Took 0.0037 seconds
hbase(main):045:0> put 'store','5','details:category','electronics'
Took 0.0044 seconds
hbase(main):046:0> put 'store','5','inventory:price','600'
Took 0.0033 seconds
hbase(main):047:0> put 'store','5','inventory:quantity','80'
Took 0.0042 seconds
hbase(main):048:0> put 'store','5','inventory:warehouse','A'
```

Q3 Scan the entire store table and confirm all 5 rows are present.

Your Answer:

```
hbase(main):049:0> scan 'store'
ROW                   COLUMN+CELL
 1                    column=details:brand, timestamp=2026-05-24T12:34:46.614, v
                      alue=dell
 1                    column=details:category, timestamp=2026-05-24T12:34:46.637
                      , value=electronics
 1                    column=details:name, timestamp=2026-05-24T12:34:46.559, va
                      lue=laptop
 1                    column=inventory:price, timestamp=2026-05-24T12:34:46.659,
                       value=1200
 1                    column=inventory:quantity, timestamp=2026-05-24T12:34:46.7
                      27, value=50
 1                    column=inventory:warehouse, timestamp=2026-05-24T12:34:46.
                      817, value=A
 2                    column=details:brand, timestamp=2026-05-24T12:34:46.883, v
                      alue=samsung
 2                    column=details:category, timestamp=2026-05-24T12:34:46.907
                      , value=electronics
 2                    column=details:name, timestamp=2026-05-24T12:34:46.857, va
                      lue=phone
 2                    column=inventory:price, timestamp=2026-05-24T12:34:46.925,
                       value=800
 2                    column=inventory:quantity, timestamp=2026-05-24T12:34:46.9
```

Q4 Retrieve all columns for the product with row key 3 (desk).

Your Answer:

```
Took 0.1169 seconds
hbase(main):050:0> get 'store','3'
COLUMN                CELL
 details:brand        timestamp=2026-05-24T12:34:47.057, value=ikea
 details:category     timestamp=2026-05-24T12:34:47.084, value=furniture
 details:name         timestamp=2026-05-24T12:34:47.036, value=desk
 inventory:price      timestamp=2026-05-24T12:34:47.102, value=300
 inventory:quantity   timestamp=2026-05-24T12:34:47.123, value=30
 inventory:warehouse  timestamp=2026-05-24T12:34:47.146, value=B
1 row(s)
Took 0.0158 seconds
hbase(main):051:0>
```

Q5 Retrieve only the inventory:price column for the product with row key 5 (tablet).

Your Answer:

```
hbase(main):051:0> get 'store','5','inventory:price'
COLUMN                CELL
 inventory:price      timestamp=2026-05-24T12:34:47.349, value=600
1 row(s)
Took 0.0076 seconds
hbase(main):052:0>
```

Q6 Use a ValueFilter scan to find all products with warehouse value A.

Hint: Use the COLUMNS and FILTER options together in your scan.

Your Answer:

```
hbase(main):001:0> scan 'store', {COLUMNS => ['inventory:warehouse'],FILTER => "
ValueFilter(=,'binary:A')"}
ROW                   COLUMN+CELL
 1                    column=inventory:warehouse, timestamp=2026-05-24T12:34:46.
                      817, value=A
 2                    column=inventory:warehouse, timestamp=2026-05-24T12:34:47.
                      001, value=A
 5                    column=inventory:warehouse, timestamp=2026-05-24T12:35:47.
                      876, value=A
3 row(s)
Took 0.7097 seconds
```

PART B — INTERMEDIATE

Questions 7–12 cover versioning, advanced filters, altering the table, and managing table lifecycle.

Q7 Enable versioning on the inventory column family to store up to 4 versions.

Your Answer:

```
Took 0.7097 seconds
hbase(main):002:0> alter 'store', NAME=>'inventory', VERSIONS=>4
Updating all regions with the new schema...
1/1 regions updated.
Done.
Took 2.0802 seconds_
```

Q8 Update the inventory:quantity for the laptop (row 1) three times: first to 45, then to 40, then to 35. Then retrieve all stored versions of that cell.

Hint: You need to alter versions first — make sure Q7 is done before this.

Your Answer:

```
hbase(main):004:0* put 'store','1','inventory:quantity','45'
Took 0.0397 seconds
hbase(main):005:0> put 'store','1','inventory:quantity','40'
Took 0.0055 seconds
hbase(main):006:0> put 'store','1','inventory:quantity','35'
Took 0.0044 seconds
hbase(main):007:0> get 'store','1',{COLUMN=>'inventory:quantity',VERSIONS=>4}
COLUMN                CELL
 inventory:quantity   timestamp=2026-05-24T12:42:31.836, value=35
 inventory:quantity   timestamp=2026-05-24T12:42:31.794, value=40
 inventory:quantity   timestamp=2026-05-24T12:42:31.741, value=45
 inventory:quantity   timestamp=2026-05-24T12:34:46.727, value=50
1 row(s)
Took 0.0238 seconds_
```

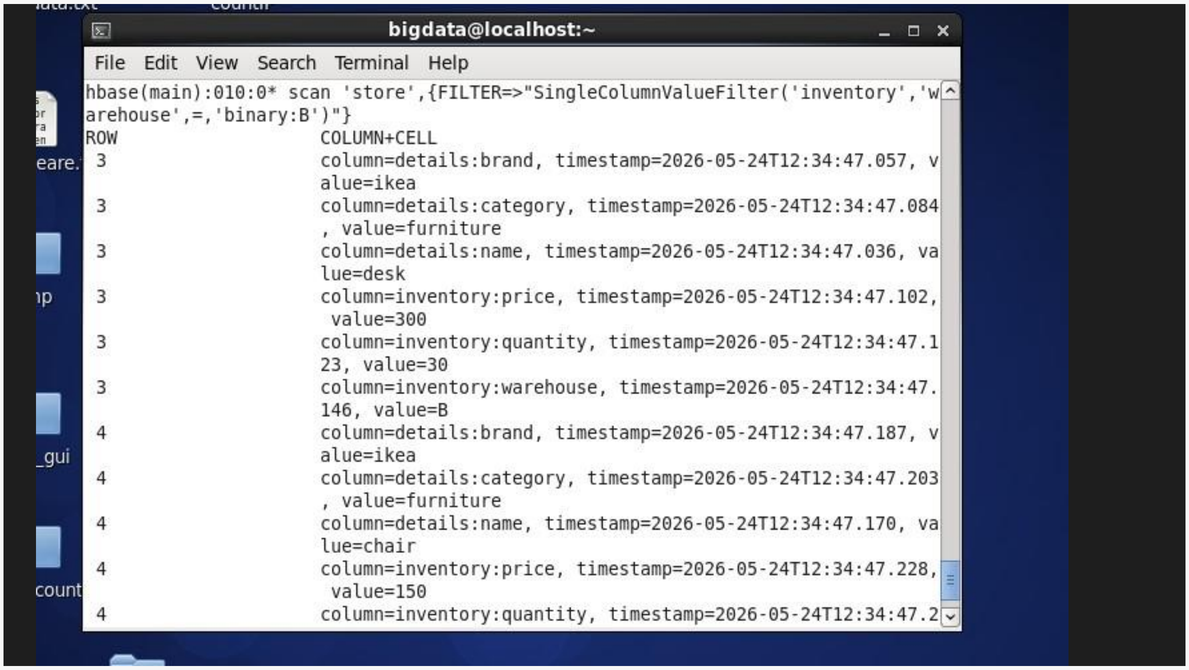
Q9 Use SingleColumnValueFilter to find all products in the furniture category.

Your Answer:

```
File  Edit  View  Search  Terminal  Help
hbase(main):008:0> scan 'store',{FILTER=>"SingleColumnValueFilter('details','cat
egory',=,'binary:furniture')"}
ROW                    COLUMN+CELL
 3                     column=details:brand, timestamp=2026-05-24T12:34:47.057, v
                       alue=ikea
 3                     column=details:category, timestamp=2026-05-24T12:34:47.084
                       , value=furniture
 3                     column=details:name, timestamp=2026-05-24T12:34:47.036, va
                       lue=desk
 3                     column=inventory:price, timestamp=2026-05-24T12:34:47.102,
                        value=300
 3                     column=inventory:quantity, timestamp=2026-05-24T12:34:47.1
                       23, value=30
 3                     column=inventory:warehouse, timestamp=2026-05-24T12:34:47.
                       146, value=B
 4                     column=details:brand, timestamp=2026-05-24T12:34:47.187, v
                       alue=ikea
 4                     column=details:category, timestamp=2026-05-24T12:34:47.203
                       , value=furniture
 4                     column=details:name, timestamp=2026-05-24T12:34:47.170, va
                       lue=chair
 4                     column=inventory:price, timestamp=2026-05-24T12:34:47.228,
                        value=150
 4                     column=inventory:quantity, timestamp=2026-05-24T12:34:47.2
```

Q10 Use SingleColumnValueFilter to find all products stored in warehouse B.

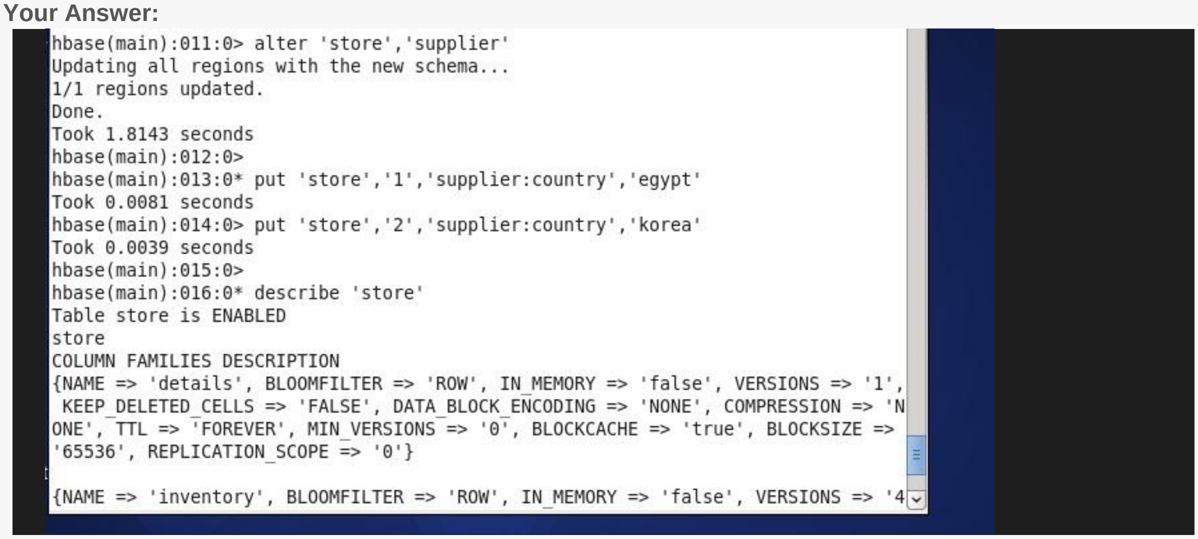
Your Answer:



```
bigdata@localhost:~
File  Edit  View  Search  Terminal  Help
hbase(main):010:0* scan 'store',{FILTER=>"SingleColumnValueFilter('inventory','w
arehouse',=,'binary:B')"}
ROW                   COLUMN+CELL
 3                    column=details:brand, timestamp=2026-05-24T12:34:47.057, v
                      alue=ikea
 3                    column=details:category, timestamp=2026-05-24T12:34:47.084
                      , value=furniture
 3                    column=details:name, timestamp=2026-05-24T12:34:47.036, va
                      lue=desk
 3                    column=inventory:price, timestamp=2026-05-24T12:34:47.102,
                       value=300
 3                    column=inventory:quantity, timestamp=2026-05-24T12:34:47.1
                      23, value=30
 3                    column=inventory:warehouse, timestamp=2026-05-24T12:34:47.
                      146, value=B
 4                    column=details:brand, timestamp=2026-05-24T12:34:47.187, v
                      alue=ikea
 4                    column=details:category, timestamp=2026-05-24T12:34:47.203
                      , value=furniture
 4                    column=details:name, timestamp=2026-05-24T12:34:47.170, va
                      lue=chair
 4                    column=inventory:price, timestamp=2026-05-24T12:34:47.228,
                       value=150
 4                    column=inventory:quantity, timestamp=2026-05-24T12:34:47.2
```

Q11 Add a new column family called supplier to the store table. Then insert supplier:country as egypt for row 1 and supplier:country as korea for row 2. Verify the schema updated correctly.

Your Answer:



```
hbase(main):011:0> alter 'store','supplier'
Updating all regions with the new schema...
1/1 regions updated.
Done.
Took 1.8143 seconds
hbase(main):012:0>
hbase(main):013:0* put 'store','1','supplier:country','egypt'
Took 0.0081 seconds
hbase(main):014:0> put 'store','2','supplier:country','korea'
Took 0.0039 seconds
hbase(main):015:0>
hbase(main):016:0* describe 'store'
Table store is ENABLED
store
COLUMN FAMILIES DESCRIPTION
{NAME => 'details', BLOOMFILTER => 'ROW', IN_MEMORY => 'false', VERSIONS => '1',
 KEEP_DELETED_CELLS => 'FALSE', DATA_BLOCK_ENCODING => 'NONE', COMPRESSION => 'N
ONE', TTL => 'FOREVER', MIN_VERSIONS => '0', BLOCKCACHE => 'true', BLOCKSIZE =>
'65536', REPLICATION_SCOPE => '0'}

{NAME => 'inventory', BLOOMFILTER => 'ROW', IN_MEMORY => 'false', VERSIONS => '4
```

```
Table store is ENABLED
store
COLUMN FAMILIES DESCRIPTION
{NAME => 'details', BLOOMFILTER => 'ROW', IN_MEMORY => 'false', VERSIONS => '1',
 KEEP_DELETED_CELLS => 'FALSE', DATA_BLOCK_ENCODING => 'NONE', COMPRESSION => 'N
ONE', TTL => 'FOREVER', MIN_VERSIONS => '0', BLOCKCACHE => 'true', BLOCKSIZE =>
'65536', REPLICATION_SCOPE => '0'}

{NAME => 'inventory', BLOOMFILTER => 'ROW', IN_MEMORY => 'false', VERSIONS => '4
', KEEP_DELETED_CELLS => 'FALSE', DATA_BLOCK_ENCODING => 'NONE', COMPRESSION =>
'NONE', TTL => 'FOREVER', MIN_VERSIONS => '0', BLOCKCACHE => 'true', BLOCKSIZE =
> '65536', REPLICATION_SCOPE => '0'}

{NAME => 'supplier', BLOOMFILTER => 'ROW', IN_MEMORY => 'false', VERSIONS => '1'
, KEEP_DELETED_CELLS => 'FALSE', DATA_BLOCK_ENCODING => 'NONE', COMPRESSION => '
NONE', TTL => 'FOREVER', MIN_VERSIONS => '0', BLOCKCACHE => 'true', BLOCKSIZE =>
 '65536', REPLICATION_SCOPE => '0'}

3 row(s)

QUOTAS
0 row(s)
Took 0.1040 seconds
```

Q12 CHALLENGE 🏆 Write a single scan that returns only products that are in the electronics category AND stored in warehouse A.

💡 *Hint: Look up FilterList with MUST_PASS_ALL to combine two SingleColumnValueFilter conditions.*

Your Answer:

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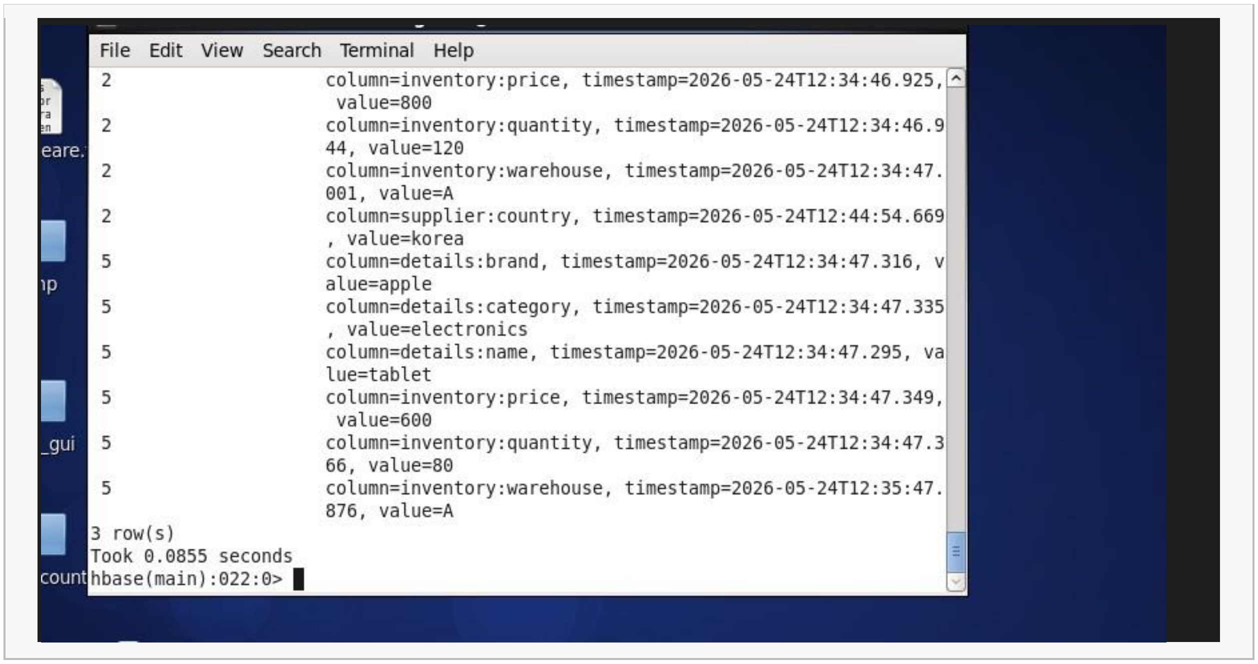
```
hbase(main):021:0> scan 'store',{FILTER =>"SingleColumnValueFilter('details','ca
tegory',=,'binary:electronics') AND SingleColumnValueFilter('inventory','warehou
se',=,'binary:A')"}
ROW                      COLUMN+CELL
 1                       column=details:brand, timestamp=2026-05-24T12:34:46.614, v
                         alue=dell
 1                       column=details:category, timestamp=2026-05-24T12:34:46.637
                         , value=electronics
 1                       column=details:name, timestamp=2026-05-24T12:34:46.559, va
                         lue=laptop
 1                       column=inventory:price, timestamp=2026-05-24T12:34:46.659,
                          value=1200
 1                       column=inventory:quantity, timestamp=2026-05-24T12:42:31.8
                         36, value=35
 1                       column=inventory:warehouse, timestamp=2026-05-24T12:34:46.
                         817, value=A
 1                       column=supplier:country, timestamp=2026-05-24T12:44:54.646
                         , value=egypt
 2                       column=details:brand, timestamp=2026-05-24T12:34:46.883, v
                         alue=samsung
 2                       column=details:category, timestamp=2026-05-24T12:34:46.907
                         , value=electronics
 2                       column=details:name, timestamp=2026-05-24T12:34:46.857, va
```



```
File  Edit  View  Search  Terminal  Help
 2                     column=inventory:price, timestamp=2026-05-24T12:34:46.925,
                        value=800
 2                     column=inventory:quantity, timestamp=2026-05-24T12:34:46.9
                       44, value=120
 2                     column=inventory:warehouse, timestamp=2026-05-24T12:34:47.
                       001, value=A
 2                     column=supplier:country, timestamp=2026-05-24T12:44:54.669
                       , value=korea
 5                     column=details:brand, timestamp=2026-05-24T12:34:47.316, v
                       alue=apple
 5                     column=details:category, timestamp=2026-05-24T12:34:47.335
                       , value=electronics
 5                     column=details:name, timestamp=2026-05-24T12:34:47.295, va
                       lue=tablet
 5                     column=inventory:price, timestamp=2026-05-24T12:34:47.349,
                        value=600
 5                     column=inventory:quantity, timestamp=2026-05-24T12:34:47.3
                       66, value=80
 5                     column=inventory:warehouse, timestamp=2026-05-24T12:35:47.
                       876, value=A
3 row(s)
Took 0.0855 seconds
hbase(main):022:0>
```

— End of Assignment —